IIT BOMBAY TECHFEST 2026-27 | NAMMASPACE CHALLENGE

High-Fidelity Monocular 3D Gaussian Splatting with Real-Time Spatial Navigation for Web-Based Digital Twins

Track: NammaSpace 3D Digital Twin Challenge (Round 1 Technical Abstract)

Live Web Application: https://nammaspace-digital-twin.vercel.app/

Source Code Repository: https://github.com/Kaish-sheikh/nammaspace-digital-twin

Abstract

Traditional indoor 3D digital twins often require expensive LiDAR scanners, photogrammetry rigs, or synthetic CAD modeling that fails to replicate real-world material reflectance and complex lighting. In this work, we present **NammaSpace**, an end-to-end framework transforming monocular smartphone video into an interactive, photorealistic 3D Gaussian Splatting (3DGS) digital twin running entirely in standard web browsers at locked 60+ FPS without local installation. By implementing a standardized **Triple-Loop capture protocol with mathematical loop closure**, our pipeline extracts dense frames with ≥75% inter-frame visual overlap, resolving sparse triangulation failures under monocular capture. Our WebGL runtime delivers a **Triple-Environment Showcase encompassing over 6,000,000+ Gaussian splats**: an Ultra-HD spatial studio twin (**1,130,000+ splats**), a grand commercial complex atrium (**1,734,000+ splats**), and an open-air campus courtyard quad (**3,135,000+ splats**). All models feature automatic mathematical ground alignment via Jacobi covariance eigendecomposition. Furthermore, we incorporate Google Maps indoor search, custom 3D landmark pins, obstacle-aware A* pathfinding, and a synchronized 2D architectural floor radar.

1. Introduction & Problem Statement

Indoor navigation across complex facilities—such as the IIT Bombay campus, airports, hospitals, and convention centers—suffers from the degradation of satellite GPS signals indoors. Existing indoor mapping solutions present severe limitations:

1. **2D Schematics:** Disorienting and non-intuitive for first-time visitors.
2. **360 Panoramas:** Rigid, discrete bubble-hopping interfaces lacking continuous 6-DoF navigation.
3. **Neural Radiance Fields (NeRFs):** Computationally prohibitive volume rendering that cannot sustain real-time frame rates in client web browsers.

NammaSpace overcomes these constraints by uniting monocular smartphone photogrammetry, real-time 3D Gaussian tile rasterization, and web-native spatial intelligence into a zero-install browser engine.

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[Monocular Video (4K, AE/AF Lock)] —> [Dense Extraction (7 FPS)] —> [Headless COLMAP (Sparse SfM)]
                                                                                    |
[Google Maps Pins & A* Navigation] <— [Three.js WebGL Engine (60 FPS)] <— [Triple-Twin Stream (6.0M+ Splats)]
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2. Capture Protocol & Standard Operating Procedure (SOP)

To eliminate tracking loss during feature matching without specialized LiDAR sensors, we developed and validated a Triple-Loop capture trajectory:

- **Loop 1 (Outer Perimeter, 0° Pitch, 1.6 m Height):** Establishes global room boundary coordinates.
- **Loop 2 (Inner Incline, 30° Pitch Down, 1.0 m Height):** Anchors ground-plane geometry and baseboards.
- **Loop 3 (Central Serpentine & 360° POI Detail Orbits):** Dense circular sweeps around key assets.
- **Exact Loop Closure:** The trajectory terminates at the exact physical starting point, allowing bundle adjustment in COLMAP to bind camera drift to <0.67 px.

3. Mathematical Formulation & Optimization

Each spatial primitive is parameterized as a 3D Gaussian ellipsoid centered at $\boldsymbol{\mu} \in \mathbb{R}^3$, with an anisotropic 3D covariance matrix $\boldsymbol{\Sigma}$:

$$\boldsymbol{\Sigma} = \mathbf{R}\mathbf{S}\mathbf{S}^T\mathbf{R}^T$$

where $\mathbf{S} = \mathrm{diag}(s_x, s_y, s_z)$ represents 3D scale factors and $\mathbf{R}$ is the rotation matrix derived from unit quaternions $\mathbf{q}$. Tile-based rasterization sorts Gaussians front-to-back across 16×16 screen tiles, blending colors via:

$$C(\mathbf{p}) = \sum_{i \in \mathcal{N}} c_i \alpha_i \prod_{j=1}^{i-1} (1 - \alpha_j)$$

where c_i is evaluated using degree-3 Spherical Harmonics to model view-dependent specular highlights, and α_i is the opacity.

Mathematical Ground-Plane Leveling

Ground alignment is computed via **Jacobi eigendecomposition** of the floor covariance matrix $\mathbf{C}_{\text{floor}} = \mathbf{V}\boldsymbol{\Lambda}\mathbf{V}^T$. The minimum eigenvector $\mathbf{v}_{\min}$ defines the floor normal, and a correction quaternion $\mathbf{q}_{\text{align}}$ rotates it to the global vertical $[0, 1, 0]^T$. This mathematically resolved a 3.54° pitch and 0.91° roll tilt in the Atrium, and a 29.18° tilt in the Courtyard.

4. WebGL Runtime Architecture & Spatial Navigation

1. **Google Maps Indoor 3D Pins:** Custom teardrop markers with billboarded labels, vertical guide beacons, and pulsing ground rings, toggled via the dock (default OFF for pure photorealism).
2. **Real-Time Search Bar:** Autocomplete filter over waypoints and 3D pins; pressing Enter smoothly reorients the camera to the destination.
3. **Obstacle-Aware A* Pathfinding:** Routes around room furniture along Catmull-Rom splines with dynamic walking duration estimates ($\approx \mathrm{Dist}/1.1\ \mathrm{m/s}$).
4. **Architectural Floor Radar:** Real-time minimap projecting player coordinates, orientation chevron, dynamic 60° FOV cone, and destination vectors.

5. Technical Specifications & Performance Benchmarks

Specification	🏛️ Flagship Studio	🏢 Commercial Atrium	🌳 Campus Courtyard
Gaussian Count	**1,130,000+ splats**	**1,734,000+ splats**	**3,135,539 splats**
Asset Format	.splat (76.8 MB)	.splat (117.8 MB)	.ksplat (75.4 MB, quantized)
Scene Type	Indoor Executive Suite	Indoor Commercial Concourse	Open-Air Quad & Terrace
Frame Rate	**60 FPS Locked**	**60 FPS Locked**	**60 FPS Locked**
Ground Alignment	Pre-calibrated (Y = 1.4 m)	Jacobi Leveled (3.54° pitch)	Leveled (29.18° pitch)

6. Conclusion & Round 3 Blueprint

NammaSpace establishes that high-fidelity spatial digital twins can be reconstructed from ordinary smartphone capture and rendered in standard web browsers at locked 60 FPS. For the Round 3 on-site challenge at IIT Bombay, our framework deploys open-source 2D Gaussian Splatting (2DGS) and Nerfstudio MCMC on provided GPU infrastructure, enforcing planar surface reconstruction on walls and floors to completely eliminate scattered pixel haze without commercial paywalls.